Unconditional Velocity-Defect Bound and the Reduction of Bad-Level Kinetic to a Homothetic-Trace Question

Abstract

We prove an unconditional pointwise inequality

$$\Big(\int_{\partial^* E_\rho} |V_\rho - \bar V_\rho|\, d\mathcal{H}^{n-1}\Big)^2 \;\le\; \Big(\frac{P(B_\rho)}{P(E_\rho)}\Big)^2 D_H(t(\rho)) \;\le\; D_H(t(\rho))$$

for every a.e. finite-perimeter torsion level $\rho \in [\rho_*, 1]$. The bound holds on *all* levels, not only the Fusco–Julin good set, and the prefactor $(P(B_\rho)/P(E_\rho))^2 \le 1$ is itself small on isoperimetrically-bad levels. Integrating against $d\mu = -t'\, d\rho$ and using the level-set deficit identity gives, with no further hypothesis,

$$\int_{\rho_*}^{\rho_\delta} \Big(\int_{\partial^* E_\rho} |V_\rho - \bar V_\rho|\, d\mathcal{H}^{n-1}\Big)^2 d\mu(\rho) \;\le\; C(n, \rho_*)\, \delta_T(\Omega).$$

The remaining obstruction to closing the bad-level kinetic action $\int_{B_I} |\dot F_\rho|^2_{\mathcal{X}} d\mu \le C\delta$ is therefore reduced to the analogous integrated bound for the *homothetic* term $\int |H_{z_\rho,\rho} - \bar V_\rho| d\mathcal{H}^{n-1}$ for some Borel centre field z_ρ defined on bad levels. We sketch two routes for this remaining piece — the centroid space–time Π-identity and a direct L^1-radial-trace construction — and identify which inputs they need.

1 Setup

Adopt the notation of `WIP_WeightedMetricTrace.tex`: torsion function $u = u_\Omega$, $-\Delta u = 1$, levels $E_\rho = \{u > t(\rho)\}$, $|E_\rho| = \omega_n \rho^n$, $w(\rho) = -t'(\rho) \ge 0$, $d\mu = w\, d\rho$, $V_\rho = w/|\nabla u|$, $\bar V_\rho = P(B_\rho)/P(E_\rho)$, D_H the Cauchy–Schwarz defect $D_H = \alpha\beta - P^2$ with $\alpha = \int 1/|\nabla u|\, d\mathcal{H}^{n-1}$, $\beta = \int |\nabla u|\, d\mathcal{H}^{n-1} = |E_\rho|$, $P = P(E_\rho)$, $P(B_\rho) = n\omega_n \rho^{n-1}$.

For convenience write $P^* := P(B_\rho)$. The fundamental flux identity $w\alpha = P^*$ (consequence of $|E_\rho| = \omega_n \rho^n$ and $\alpha = -m'(t(\rho))$, $w = -t'(\rho)$) is used throughout.

2 The unconditional velocity-defect bound

Lemma 2.1 (Weighted variance of V)**.** *For a.e. $\rho \in [\rho_*, 1]$ at which E_ρ has finite perimeter,*

$$\int_{\partial^* E_\rho} \frac{(V_\rho - \bar V_\rho)^2}{V_\rho}\, d\mathcal{H}^{n-1} \;=\; \frac{P^*}{P^2}\, D_H(t(\rho)). \tag{1}$$

Proof. Set $f := V_\rho = w/|\nabla u|$, $\bar f := \bar V_\rho = P^*/P$. Expanding $(f - \bar f)^2/f = f - 2\bar f + \bar f^2/f$ and integrating,

$$\int (f - \bar f)^2/f = \int f - 2\bar f P + \bar f^2 \int 1/f.$$

Now $\int f = \int V_\rho = w\alpha = P^*$ (flux); $\int 1/f = \int |\nabla u|/w = \beta/w$; $\bar{f}P = P^*$ and $\bar{f}^2 = P^{*2}/P^2$. Hence

$$\int (f-\bar{f})^2/f = P^* - 2P^* + \frac{P^{*2}}{P^2}\cdot\frac{\beta}{w} = -P^* + \frac{P^*}{P^2}\cdot\frac{P^*\beta}{w} = -P^* + \frac{P^*}{P^2}\,\alpha\beta,$$

where the last equality used $w\alpha = P^*$ to write $P^*\beta/w = \alpha\beta$. Therefore

$$\int (f-\bar{f})^2/f = \frac{P^*}{P^2}\big(\alpha\beta - P^2\big) = \frac{P^*}{P^2}\,D_H. \qquad \square$$

Theorem 2.2 (Unconditional L^1 velocity-defect bound)**.** *For a.e. $\rho \in [\rho_*, 1]$ at which E_ρ has finite perimeter,*

$$\Big(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho|\,d\mathcal{H}^{n-1}\Big)^2 \;\le\; \Big(\frac{P^*}{P}\Big)^2 D_H(t(\rho)) \;\le\; D_H(t(\rho)). \tag{2}$$

No isoperimetric smallness or Fusco–Julin good-level assumption is used.

Proof. By Cauchy–Schwarz in the form $\big(\int |g|\,d\mathcal{H}^{n-1}\big)^2 \le \big(\int g^2/V_\rho\,d\mathcal{H}^{n-1}\big)\big(\int V_\rho\,d\mathcal{H}^{n-1}\big)$ applied to $g := V_\rho - \bar{V}_\rho$ (with the standard $\sqrt{V_\rho}\cdot\sqrt{1/V_\rho}$-factorisation),

$$\Big(\int |V_\rho - \bar{V}_\rho|\Big)^2 \le \Big(\int \frac{(V_\rho - \bar{V}_\rho)^2}{V_\rho}\Big)\Big(\int V_\rho\Big) = \frac{P^*}{P^2}\,D_H\cdot P^* = \Big(\frac{P^*}{P}\Big)^2 D_H,$$

using Lemma 2.1 and $\int V_\rho = P^*$. The second inequality of (2) follows from $P \ge P^*$ (isoperimetric inequality). $\square$

Remark 2.3 (Strengthening over TrimmedVelocityRepair)**.** The estimate (2) is sharper than the bound proved in `WIP_TrimmedVelocityRepair.tex` in two ways.

1. It is *unconditional*: the previous argument restricted to the Fusco–Julin good set $G_I = \{D_I \le \eta_I\}$ via a truncation $\{f \ge \bar{f}/2\}$. No such restriction is needed here. The Cauchy–Schwarz $(\int |g|)^2 \le (\int g^2/V)(\int V)$ does not require a lower bound on V: it weights the variance by V itself, making the integrand integrable wherever V is.
2. The prefactor $(P^*/P)^2 \le 1$ is itself *smaller* on isoperimetrically bad levels. Since $\mathcal{I}(\rho) = P^2 - P^{*2} = (P-P^*)(P+P^*)$, we have $(P^*/P)^2 = 1 - \mathcal{I}/P^2$. In the regime where $P \gg P^*$ on a bad level, the velocity defect is suppressed by a factor $\le (P^*/P)^2 \le P^{*2}/(P^{*2}+\mathcal{I})$.

Corollary 2.4 (Integrated velocity-defect on all levels)**.**

$$\int_{\rho_*}^{\rho_\delta} \Big(\int_{\partial^* E_\rho} |V_\rho - \bar{V}_\rho|\,d\mathcal{H}^{n-1}\Big)^2 d\mu(\rho) \;\le\; \int_{\rho_*}^{\rho_\delta} D_H(t(\rho))\,d\mu(\rho) \;\le\; C(n,\rho_*)\,\delta_T(\Omega). \tag{3}$$

Proof. Apply Theorem 2.2 pointwise, then integrate; the second inequality is the level-set deficit identity of `WIP_LevelSetIdentity.tex`. $\square$

3 Reduction of the bad-level kinetic problem

Combining (2) with the unconditional metric upper gradient of `WIP_LimsupDeformation.tex`,

$$|\dot{F}_\rho|_{\mathcal{X}} \le \rho^{-n} \inf_{a\in\mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho,\rho} - a\cdot\nu_\rho|\,d\mathcal{H}^{n-1},$$

specialising to $a = 0$, and the triangle inequality $|V_\rho - H_{z_\rho,\rho}| \le |V_\rho - \bar V_\rho| + |H_{z_\rho,\rho} - \bar V_\rho|$, we obtain pointwise

$$|\dot F_\rho|^2_{\mathcal X} \;\le\; 2\rho^{-2n}\Big[D_H(t(\rho)) + \big(\int |H_{z_\rho,\rho} - \bar V_\rho|\, d\mathcal H^{n-1}\big)^2\Big] \tag{4}$$

for any bounded Borel centre field z_ρ. This holds on *every* a.e. finite-perimeter level, with no FJ smallness.

Proposition 3.1 (Reduction to homothetic-trace)**.** *Suppose there is a bounded Borel centre field $\rho \mapsto z_\rho \in \mathbb R^n$, $|z_\rho| \le R'$, and a constant $C_{\rm hom} = C_{\rm hom}(n, R, R', \rho_*) > 0$ such that*

$$\int_{\rho_*}^{\rho_\delta} \Big(\int_{\partial^* E_\rho} |H_{z_\rho,\rho} - \bar V_\rho|\, d\mathcal H^{n-1}\Big)^2 d\mu(\rho) \;\le\; C_{\rm hom}\, \delta_T(\Omega). \tag{5}$$

Then

$$\int_{\rho_*}^{\rho_\delta} |\dot F_\rho|^2_{\mathcal X}\, d\mu(\rho) \;\le\; C(n, R, R', \rho_*)\, \delta_T(\Omega). \tag{6}$$

In particular the weighted bad-isoperimetric action (eq:bad-I-action) *of `WIP_WeightedMetricTrace.tex` is discharged after restriction to B_I. This does* not *discharge the unweighted bad-density action* (eq:bad-tau-action)*, which is a $d\rho$-estimate on $\{w < \tau_0\}$.*

Proof. Integrate (4) against $d\mu$ and use Corollary 2.4 for the first piece, (5) for the second, and $\rho^{-2n} \le \rho_*^{-2n}$ on $[\rho_*, \rho_\delta]$. Restricting (6) to B_I gives the bad-I action. On $G_\tau = \{w \ge \tau_0\}$, the same estimate also implies the corresponding $d\rho$-kinetic bound after division by τ_0. No conclusion follows on $B_\tau = \{w < \tau_0\}$, because $d\mu = w\, d\rho$ may be arbitrarily small there. $\square$

4 Status of (5): two candidate routes

The conditional input (5) is the *only* remaining input needed to make Plan 2 / Route δ unconditional. Two routes for proving it are visible.

4.1 Route I: centroid space–time Π-identity

Choose $z_\rho := \bar z_\rho$, the centroid of E_ρ. The divergence identity [4, Eq. (5.3′)],

$$\int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{z_\rho}|^2 d\mathcal H^{n-1} = 2\big(P(E_\rho) - P(B_\rho)\big) + 2\Pi(E_\rho, \bar z_\rho),$$

combined with the radial trace estimate of [5, Thm. 2.2], gives

$$\Big(\int |H_{\bar z_\rho,\rho} - \bar V_\rho|\, d\mathcal H^{n-1}\Big)^2 \le C(n, R, \rho_*)\Big[|E_\rho \Delta B_\rho(\bar z_\rho)|^2 + \big(\int |\nu - e_{z_\rho}|^2 d\mathcal H^{n-1}\big)^2\Big]. \tag{7}$$

By the pointwise slice-rearrangement inequality $|E_\rho \Delta B_\rho(\bar z_\rho)|^2 \le C\,\Pi(E_\rho, \bar z_\rho)$ of [4, Lem. 4.3] and the integrated identity

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar z_\rho)\, d\mu \le C_\Pi\, \delta_T \qquad (\Pi\text{-int})$$

of the same block, *plus* an integrated bound on $\big(\int |\nu - e_{z_\rho}|^2\big)^2$ which is open (naive Cauchy–Schwarz only yields a $\sqrt{\delta_T}$ rate, not δ_T), one obtains (5). The remaining sub-target inside Route I is therefore

$$\int_{\rho_*}^{\rho_\delta} \Big(\int_{\partial^* E_\rho} |\nu - e_{z_\rho}|^2\, d\mathcal H^{n-1}\Big)^2 d\mu(\rho) \;\le\; C\, \delta_T,$$

which is currently isolated as a separate open problem.

4.2 Route II: oriented radial flux on bad levels

The oriented divergence trace [5, Thm. 2.1] gives, for *any* centre z with $|z| \le R+1$,

$$\int_{\partial^* E_\rho} \Big|\tfrac{|x-z|}{\rho} - 1\Big| d\mathcal{H}^{n-1} \le \tfrac{n}{\rho_*}|E_\rho \Delta B_\rho(z)| + \tfrac{W(R,\rho_*)}{2} \int |\nu - e_z|^2 d\mathcal{H}^{n-1},$$

unconditionally on the deficit. If z is chosen as the *Fraenkel* centre $z_F = z_F(E_\rho)$, then $|E_\rho \Delta B_\rho(z_F)|^2 \le C_F\,\mathcal{I}(\rho)$ holds unconditionally (Figalli–Maggi–Pratelli). The squared integrated $\int |\nu - e_{z_F}|^2 d\mathcal{H}^{n-1}$ at the *Fraenkel* centre is not $\le C\mathcal{I}$ in general — that is the input that fails when one tries to use the Fraenkel centre as a substitute for the FJ centre on bad levels. Closing Route II therefore reduces to bounding $\int \big(\int |\nu - e_{z_F}|^2\big)^2 d\mu$ separately, with the same kind of structural difficulty as Route I.

5 Summary of progress

- Theorem 2.2 is new: the L^1-velocity defect $\int |V - \bar{V}| d\mathcal{H}^{n-1}$ is controlled by $\sqrt{D_H}$ on *every* finite-perimeter level, with prefactor $P^*/P \le 1$. The previous TrimmedVelocityRepair bound was good-level only.
- Corollary 2.4 integrates this against $d\mu$ unconditionally, giving the half of the kinetic action that involves D_H directly.
- Proposition 3.1 reduces the weighted kinetic problem, and hence the bad-I action, to the single integrated homothetic-trace bound (5). It does not remove the separate bad-τ $d\rho$-action requirement.
- Section 4 isolates the residual difficulty as

  $$\int_{\rho_*}^{\rho_\delta} \Big(\int |\nu - e_{z_\rho}|^2 d\mathcal{H}^{n-1}\Big)^2 d\mu(\rho) \;\le\; C\,\delta_T,$$

  for either the centroid centre $z_\rho = \bar{z}_\rho$ (Route I, via II-int) or the Fraenkel centre $z_\rho = z_F$ (Route II, direct). This is a clean, decoupled mathematical question and not specific to Fusco–Julin.

References

[1] Plan 2 building block LevelSetIdentity, `WIP_LevelSetIdentity.tex`.

[2] Plan 2 Deliverable A, LimsupDeformation, `WIP_LimsupDeformation.tex`.

[3] Plan 2 building block TrimmedVelocityRepair, `WIP_TrimmedVelocityRepair.tex`.

[4] Plan 2 auxiliary block SpaceTimeIdentity, `WIP_SpaceTimeIdentity.tex`.

[5] Plan 2 bridge block FJCenterBridge, `WIP_FJCenterBridge.tex`.

[6] Plan 2 building block WeightedMetricTrace, `WIP_WeightedMetricTrace.tex`.